

- 2 Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are 58° and 47° . Find the distance from the balloon to the first observer (the one with the 58° sighting), to the nearest metre.

- 3 In triangle PQR , two sides of length 8 and 11 enclose an angle of 52° . Find the length of the third side, to two decimal places.
- 4 A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.
- 5 Find the area of a triangle with sides 10 and 14 enclosing an angle of 35° , to two decimal places.
- 6 A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N 50^\circ E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

- 7 In triangle ABC , $A = 30^\circ$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B .
- 8 In triangle DEF , $D = 40^\circ$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

Choosing the right law

- 9** For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:
- a** Two angles and one side are known (AAS).
 - b** Three sides are known (SSS).
 - c** Two sides and the included angle are known (SAS).
- 10** A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.